

MINH SON LE

info.leminhson@gmail.com | (859) 786 5738 | github.com/DaRealSun | linkedin.com/in/son leminh

Washington DC Baltimore Area

SUMMARY

Oracle Certified Professional (Java SE 11) and Full-Stack Engineer with over 4 years of experience specializing in robust **Spring Boot** ecosystems, cloud-native microservices, and cutting-edge **Agentic AI** orchestration. Proven track record of designing highly scalable, event-driven architectures utilizing Apache Kafka and AWS, with a sharp expertise in data-layer optimization that has successfully delivered **10x improvements** in service response times. Adept at bridging enterprise Java backend systems with next-gen AI engineering—actively architecting intelligent, retrieval-augmented generation (RAG) routing workflows using **Spring AI**, **LangChain4j**, and **Model Context Protocol (MCP)** tools.

CERTIFICATIONS

- [Oracle Certified Professional, Java SE 11 Developer](#) Issued February 2026. Exam Score: 86 percent.
- [Java Certified Foundations Associate](#) Issued November 2025.

EXPERIENCE

Software Engineer

July 2022 – October 2025

FPT Software – Contract Remote Vietnam

- **Agentic AI document routing:** Designed an **agentic** workflow that classifies and routes cases via **Retrieval Augmented Generation** over a **pgvector** index (**Spring AI**, **LangChain4j**) using **Model Context Protocol (MCP)** tools.
- **Backend services & microservices:** Using **Spring Boot** and **AWS Lambda**, improving system responsiveness by 30% during peak load. Owned a claim intake service end to end **REST API** contracts, **PostgreSQL** schema, and **Apache Kafka** topic layout for downstream consumers within an **event driven architecture**.
- **Cloud infrastructure:** Implemented **AWS Elastic Load Balancing** and **IAM** roles for **fault tolerant**, **cloud native** deployment, ensuring 90% system uptime and robust resource authorization; containerized services with **Docker** and orchestrated them on **Kubernetes**.
- **Concurrency & throughput:** Optimized data processing throughput by 70% by architecting a concurrent system with **ExecutorService** and **Callable** on custom thread pools for **asynchronous** task execution, then benchmarked and migrated hot paths to **Java Virtual Threads** using **JMH**.
- **JVM & data performance:** Analyzed and eliminated slow system performance stemming from **N+1 query patterns** by implementing **Hibernate fetch joins** and optimized **SQL queries** with pagination and lazy loading, improving service response times by **10x**.
- **Enterprise Security:** Implemented **OAuth 2.0** and **JWT-based Single Sign-On (SSO)** across 3 internal applications using **Spring Security**, effectively mitigating unauthorized API access.
- **Data Layer & Observability:** Managed complex relational datasets using **Spring Boot** and **Spring Data JPA**; reduced data retrieval latency by **40%** by integrating **Amazon DynamoDB** and streamlined object-mapped data access with **AWS CloudWatch** for proactive bottleneck detection.
- **Test Automation:** Built and enhanced a reusable test automation framework covering **unit** (JUnit 5, Mockito), **API** (REST-assured, Postman), **integration**, and **UI** (Selenium) layers.
- **CI/CD & DevOps:** Integrated automated testing suites into **Jenkins** CI/CD pipelines utilizing **Docker** containerization and **GitHub** source control.
- **SDLC:** Executed the full **SDLC within Agile/Scrum** environments, driving weekly code reviews, authoring onboarding guides for new Proof of Concepts (PoCs), and partnering with security and non-technical stakeholders to deliver shippable software.
- **Architected and implemented RESTful Web Services** using both **Spring Boot** and **JAX-RS** interfaces, designing endpoints that produce/consume **JSON** and **XML** payloads to ensure seamless cross-system integration and code reusability.

- **Developed Spring MVC framework components**, utilizing **Spring IoC containers** for clean dependency injection across Controllers, Services, and Repositories to maintain a decoupled architecture.
- **Validated system reliability** by performing automated and manual endpoint testing using **Postman** to guarantee comprehensive API functional correctness.
- **Engineered a robust DAO layer** utilizing **Spring Data JPA repositories** and the **Hibernate** framework, managing complex data models and invoking optimized database stored procedures for complex queries.
- **Integrated observability metrics** with method-level granularity across backend services using **Spring Actuator** to track and optimize system performance.
- **Designed and managed cloud infrastructure on AWS**, leveraging **EC2**, **RDS**, **S3 buckets**, **Route 53**, **AWS Lambda functions**, and **SNS Topics** to build highly available backend systems.
- **Ensured high availability and fault tolerance** across cloud environments by implementing **AWS Elastic Load Balancing** and **Auto Scaling** groups.
- **Optimized the DevOps lifecycle** by managing source code within **Git**, tracking project deliverables in **Jira**, and automating continuous integration pipelines using **Maven** and **Jenkins**.

Web Developer

September 2021 – April 2022

FPT Software – Contract Remote Vietnam

- **Customer Portal (SPA)**: Engineered modular frontend components and enterprise routing using **React** and **Angular** for a member-facing single-page application serving 10,000 monthly users within a multi-developer Agile team.
- **State & Performance Optimization**: Evaluated the Context API and implemented **Redux Toolkit (with async thunks)** to design predictable global state containers and actions; boosted **Lighthouse performance scores from 72 to 94** via code splitting and route-level lazy loading.
- **Angular Enterprise Patterns**: Mastered framework lifecycles (`ngOnInit`, `ngOnChanges`, `ngOnDestroy`) to manage asynchronous HTTP data streams and unsubscribe from **RxJS observables** to eliminate memory leaks; optimized rendering with `trackBy` structural directives.
- **Secure Routing & Forms**: Enforced strict application authorization across endpoints by implementing advanced Angular **route guards** (`CanActivate`, `CanLoad`, `Resolve`); designed complex, type-safe **reactive and template-driven forms** utilizing dependency injection.
- **Responsive UI & Accessibility**: Built mobile-first, responsive layouts utilizing **HTML5**, **CSS3 Flexbox/Grid**, and **Bootstrap** while maintaining strict compliance with **WCAG 2.1 AA accessibility standards** via semantic markup, ARIA roles, and full keyboard navigation.
- **Interactive Data Visualization**: Integrated the **Google Maps API** and **YouTube API** to build highly interactive data presentation layers, deploying caching and throttling strategies to efficiently manage API quotas.
- **Node.js Services**: Developed **Node.js** services to serve as a lightweight Backend-for-Frontend (BFF) layer, exposing optimized REST endpoints explicitly tailored for SPA consumption.
- **Automated Testing**: Authored comprehensive unit and integration test suites using **Jest** and **React Testing Library**, consistently achieving **90%+ code coverage** and eliminating regressions before QA delivery.

PROJECTS

Live Facial Emotion Recognition [Northern Kentucky University](#), 2024

Python, Tensorflow, RAG, Agent to Agent

Research conducted under Dr. Junxiu Zhou, Associate Professor of Computer Science at NKU, specializing in machine learning, deep learning, and computer vision. Designed and trained a custom 5 layer Convolutional Neural Network in TensorFlow and Keras for real time recognition of seven facial emotions, trained on the [FER2013 dataset](#).

TECHNICAL SKILLS

Languages: Java (11, 17, 21), Python, Go, C++, TypeScript, JavaScript, Python, SQL

Backend & APIs: Spring Boot, Spring Data JPA, Spring Security, Spring AI, Hibernate, REST APIs, GraphQL, Microservices, Event Driven Architecture, Apache Kafka, PostgreSQL, Amazon DynamoDB, Redis

Testing & QA Automation: JUnit 5, Mockito, REST assured, Testcontainers, Selenium, Jest, React Testing Library; unit / API / integration / UI automation; CI integrated test suites

Concurrency & JVM: Virtual Threads (Project Loom), ExecutorService, Callable, CompletableFuture, JDK Flight Recorder, GC tuning, JMH benchmarking

AI & Machine Learning: Agentic workflows, Model Context Protocol (MCP), LangChain4j, pgvector, embedding based retrieval, Retrieval Augmented Generation, TensorFlow, Keras, Convolutional Neural Networks, MediaPipe

Cloud & DevOps: AWS (Lambda, ELB, IAM, CloudWatch, DynamoDB), Google Cloud, Docker, Kubernetes, Jenkins, GitHub Actions, Maven, Gradle, CI/CD

Frontend: Angular, React, Redux Toolkit, React Router, Tailwind CSS, Node.js

Observability: OpenTelemetry, Jaeger, Micrometer, AWS CloudWatch

Practices: Agile/Scrum, SDLC, cloud native & fault tolerant design, secure API design (OAuth 2.0, JWT, Bcrypt)

EDUCATION

[Northern Kentucky University](#) Graduated May 2025

Bachelor of Science in Computer Science, Minor in Mathematics Cum Laude

Awards and Honors:

- International Merit Scholarship Northern Kentucky University
- EDGE Award Northern Kentucky University
- Dr. Charles Frank Scholarship Northern Kentucky University
- Sehnert Pugh Scholarship Northern Kentucky University